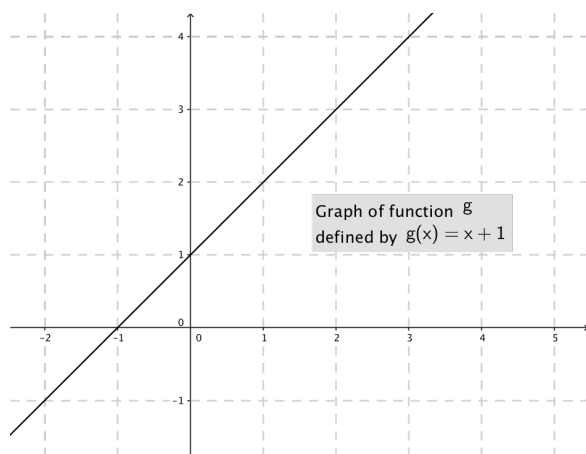
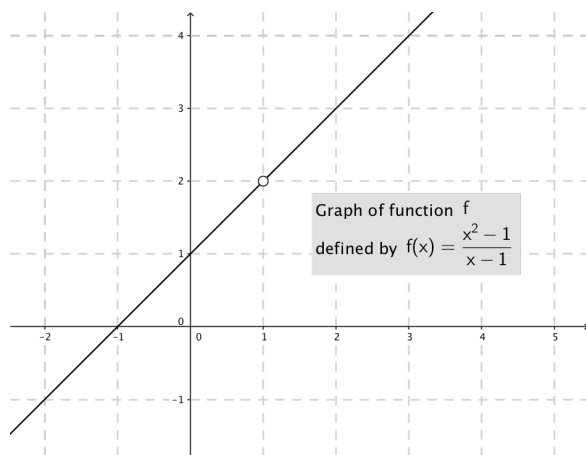


Problem 2

Problem 2

Problem 1 Use the graphs and the given definitions of the following two functions to answer the questions below.



(a) Find the domain of f and the domain of g .

Problem 2

Explanation. The domain of f is $(-\infty, 1) \cup (1, \infty)$ (all real numbers except 1). The domain of g is $(-\infty, \infty)$ (all real numbers).

(b) *Is $f = g$? (Why or why not?)*

Explanation. No, these two functions are not equal. Two functions are equal if and only if they have identical domains and their values agree on all points in the domain.

Since f and g have different domains, by part (a), they cannot be equal functions.

(c) *Looking at the graphs, find $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow 1} g(x)$.*

Explanation. From the graphs we have that $\lim_{x \rightarrow 1} f(x) = 2$ and $\lim_{x \rightarrow 1} g(x) = 2$.